

# Abhinav Kataria

Backend-Leaning Full-Stack Engineer | Systems & AI Integration

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## SUMMARY

Final year Computer Science Engineering student at **BIT Mesra** who builds production-style, real-time systems - from architecting an enterprise LMS's **authorization and privacy pipeline** solo, to contributing to **NexRoute**, a live-deployed, team-built delivery-allocation platform. Backed by strong **DSA** fundamentals (500+ problems solved, **rank 1769 in LeetCode Biweekly 180**) and experience mentoring junior developers as IEEE Technical Lead.

## EDUCATION

**Birla Institute of Technology, Mesra**

2023 - 2027

Bachelor of Engineering in Computer Science Engineering

- **CGPA:** 7.5/10
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

**Prior:** *Vishwa Bharati Public School, Noida (2008 – 2022) - Elected Vice Head Boy (2017-2018); ranked top 5% of batch*

## TECHNICAL SKILLS

**Languages:** Java, C++, C, JavaScript, SQL

**Backend & Databases:** Node.js, Express, MongoDB, Mongoose, FastAPI, REST APIs, Socket.IO/WebSockets

**Frontend:** React, Next.js, Mapbox GL, Tailwind CSS, Shadcn UI, HTML/CSS

**Auth, Security & Systems:** JWT, OAuth 2.0, Role-Based Access Control, Rate Limiting, Geospatial Indexing (H3)

**Core Concepts:** DSA, OOPs, DBMS, OS, CN, System Design, CI/CD, Git/GitHub

## WORK EXPERIENCE

**IT Intern, Jakson Limited**

June 2026 - Present

- Built an Authorization Engine enforcing role-based access across 4 enterprise roles (Learner, Manager, HR/Admin, Support), eliminating unauthorized cross-role data exposure by design.
- Engineered a dual-stage privacy pipeline that strips PII (emails, employee IDs, session tokens) before and after every LLM call, closing two of the most common enterprise-AI leak vectors.
- Cut unnecessary LLM calls by routing FAQ and navigation queries through a deterministic Knowledge Retrieval Engine, reserving the LLM layer for genuinely ambiguous requests only.
- Hardened the conversational interface against real-world input noise (typo tolerance, synonym matching, ranked retrieval) inside a FastAPI-based backend pipeline.

## PROJECTS

**NexRoute - Real-Time Delivery Partner Allocation Engine**

- Contributed across the stack of a team-built, real-time delivery-allocation platform (React, Node/Express, Socket.IO, MongoDB) simulating live rider dispatch over a deterministic, H3-geospatial allocation engine - with a focus on frontend architecture and bug fixing.
- Found and fixed a live authorization bug that defaulted every OAuth-created account to admin-level access, correcting the role assignment and verifying the fix directly against production data.
- Diagnosed and fixed a race condition between the app's WebSocket connection and login state that caused false "disconnected" errors, by rebuilding the socket lifecycle to track authentication state instead of firing on initial app mount.
- Built the map-based location-picker component (click-to-set coordinates synced with manual input) and resolved a REST/WebSocket field-naming mismatch that was breaking live rider-position rendering.

## POSITION OF RESPONSIBILITY

**Technical Lead, IEEE Student Chapter, BIT Mesra**

Nov 2024 - Present

Mentored junior developers in web development and coding-competition prep; helped organize hackathons and technical workshops.

**Technical Member, GDSC BIT Mesra**

Oct 2023 - Present

Contributed to full-stack team projects; ran peer coding sessions.

## ACHIEVEMENTS

- **Technical:** Qualified JEE Main & Advanced 2023 - top 4% (category); solved 500+ coding problems across competitive programming platforms; rank **1769** in LeetCode Biweekly Contest 180 (All Killed)
- **Academic:** Cleared NTSE Level 1 (2019); scored **95%** in CBSE Class 10 Board Examinations